

Assignment 4.1 Setting Up Ansible for Server Management	
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Part 1: Research

What is Ansible and why is it widely used in server management?

- Ansible is a tool that helps manage and control many servers at once. It is used to install softwares, update systems, and configure servers automatically. It saves time in a way we do not have to install or update systems in multiple servers one by one, instead, we use ansible to control them all at once.

What role does SSH play in how Ansible manages servers?

- SSH lets Ansible securely connect to servers. Ansible uses SSH to send commands and tasks to servers without needing extra software installed on them.

What are the purposes of the following files?

- `ansible.cfg`
 - This file stores Ansible's settings, like default paths, user names, and connection options.
- `inventory.ini`
 - This file lists all the servers that Ansible will manage and groups them if needed.
- A playbook (.yml file)
 - This file contains the tasks and instructions that Ansible will run on the servers, written in simple steps.

Part 2: Reading & Summary

Read about how to install Ansible on Ubuntu from at least one reliable online source (official docs, blogs, tutorials).

Write a short summary (5–7 sentences) explaining:

- **The installation steps**
- **The purpose of an inventory**
- **The purpose of a configuration file**

To install Ansible on Ubuntu, we need to update the package list and add the official Ansible repository (PPA). After that, we refresh the package index again and install Ansible using the `apt install ansible` command. Once installed, we can check that it worked by running `ansible --version` (Heidi, 2022). An inventory is a file where we list all the servers (hosts) that Ansible

will manage. It can group servers and tell Ansible which machines to talk to (Heidi, 2022). A configuration file (ansible.cfg) is where we set options that control how Ansible behaves, like where to find the inventory, default user names, and connection settings (Unix Tutorial, 2019).

Part 3: Organizing

Create a simple draft version of:

- A sample inventory.ini (with at least 2 pretend server IPs).
- A sample ansible.cfg file with at least 2 meaningful settings (e.g., inventory and remote_user).

None

```
#inventory.ini
```

```
[managed_nodes]
```

```
192.168.56.101
```

```
192.168.56.102
```

None

```
#ansible.cfg
```

```
[defaults]
```

```
inventory = inventory.ini
```

```
remote_user = ubuntu
```

```
host_key_checking = False
```

Part 4: Install a useful application used in Development, do not use applications that we have installed before.